Statistical Inference and Modeling

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Lectures

- Lecture 1: Introduction
- Lecture 2: Model selection
- Lecture 3: Resampling
- Lecture 4: Linear regression
- Lecture 5: Robust regression
- Lecture 6: Classification
- Lecture 7: Logistic regression
- Lecture 8: Neural networks
- Lecture 9: K-nearest neighbor
- Lecture 10: Bayesian decision, LDA, QDA, and naive Bayes
- Lecture 11: Support vector machine
- Lecture 12: Tree-based methods
- Lecture 13: Bagging, random forecast, boosting
- Lecture 14: Unsupervised learning and clustering algorithms
- Lecture 15: Unsupervised learning and PCA
- Lecture 16: Gaussian mixture model clustering
- Lecture 17: DBSCAN clustering

7. Classification

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1 Table of Contents

2 Classification

3 Binary classification performance
- Standard classification indices
- Trade-off in classification
- Receiver operating characteristic (ROC)
- Precision-recall curve

4 Accuracy, precision, and recall in multi-class classification
- Accuracy
- Problem with accuracy
- Precision, recall, F1 per class
- Micro, macro, and weighted-averaging

What is a classification?

Classification is a process that assigns the observation to a class

- Often, a method first predicts the probability of a class of a qualitative variable; or it provides rules to assign observations to a class;
- A classification technique is called **classifier**.

Example

A classification of patients heart disease condition: mild, moderate, and severe (3 classes) using $\boldsymbol{x} :=$ [LDL, BMI, alcohol, age] as predictors.

A classifier can provide threshold-based rules on x:

- $\hat{y} =$ mild, if LDL < 40 and BMI < 30.
- $\hat{y} =$ moderate, if (LDL $\in [100, 120]$ and alcohol > 10) or (LDL $\in [120, 200]$ and age > 40) or (BMI $\cdot$ alcohol $\in [1, 100]$).
- $\hat{y} =$ severe, uf (LDL > 200) or (BMI $\cdot$ alcohol > 200).

A classifier approximate the probability of each class (given an input features $\boldsymbol{x}$) via mathematical functions. A classifier maps the input $\boldsymbol{x}$ to a probabilistic value (0-1), $\hat{f}(\cdot) : \mathcal{R}^n \Rightarrow [0, 1]^C$:

$$p(Y = c|\boldsymbol{x}) = \hat{f}(y_c|\boldsymbol{x}, \theta)$$

where θ is the model parameter needed to be identified.

Settings.

- **Training process:** involves how to find good thresholds or the best parameter θ, using training data.
- **Testing process:** given a new value of $\boldsymbol{x}$, we predict $\hat{y}$ using the estimated parameter θ.

Why not linear regression?

Suppose we try to predict a medical condition of three cases: stroke, drug overdose, epileptic seizure.

$$\text{Choice 1:} \quad \hat{y} = \begin{cases} 0, & \text{normal} \\ 1, & \text{stroke} \\ 2, & \text{drug overdose} \\ 3, & \text{epileptic seizure} \end{cases} \qquad \text{Choice 2:} \quad \hat{y} = \begin{cases} 0, & \text{normal} \\ 1, & \text{epileptic seizure} \\ 2, & \text{stroke} \\ 3, & \text{drug overdose} \end{cases}$$

- If you use a linear regression, *i.e.*, $\hat{y} = \boldsymbol{x}_{\text{Test}}^T \boldsymbol{\beta}$, then the value of $\hat{y}$ could be thresholded into 4 categories, just like in Choice 1 and 2.
- However, the value of $\hat{y} = \boldsymbol{x}_{\text{Test}}^T \boldsymbol{\beta}$ cannot catch up with the lack of natural ordering in these labels.
- The different distance between predictions, *e.g.*, 0 vs 1 or 0 vs 2 should be the same. Thus, the **first reason** is the mapping value in linear regression cannot reflect the unnatural ordering of the labels.

Why not linear regression?

If there are only two possibilities: stroke or drug overdose.

$$\text{Choice 1:} \quad y = \begin{cases} 0, & \text{stroke} \\ 1, & \text{drug overdose} \end{cases} \qquad \text{Choice 2:} \quad \tilde{y} = \begin{cases} 0, & \text{drug overdose} \\ 1, & \text{stroke} \end{cases}$$

- We can fit a linear regression using the first choice of encoding and predict drug overdose if $y > 0.5$ (or for choice 2, predict stroke if $\tilde{y} > 0.5$).
- However, if we use linear regression, some of our estimates might be outside the [0, 1] interval, making them hard to interpret as probabilities!
- **Second reason:** as we know in linear regression $\hat{y} = \boldsymbol{x}_{\text{Test}}^T \boldsymbol{\beta}$, some of our estimates might be outside [0, 1]. So, we need a better mapping function to ensure that $\hat{y}$ can return the predicting probability.

Discriminative vs. generative approach for classifiers

Discriminative approach:

- Learn the parameters from $p(Y = c|\boldsymbol{X})$, directly.

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\textbf{argmax}}\ p(Y = c|\boldsymbol{X}, \boldsymbol{\beta})$$

- Learn mappings from inputs to classes (least-squares, decision trees).
- *e.g.*logistic regression, KNN, NN classifier, SVM, decision trees.

Generative approach:

- Learn the parameters from $p(\boldsymbol{X}|Y)$ or $p(\boldsymbol{X}, Y)$ or $p(\boldsymbol{X})$

$$\boldsymbol{\beta}^* = \underset{\boldsymbol{\beta}}{\textbf{argmax}}\ \log p(\boldsymbol{X},\ \boldsymbol{\beta}, Y = c) = \log \frac{p(\boldsymbol{X}, \boldsymbol{\beta}|Y = c)}{p(Y = c)}$$

- We then use Bayes' theorem to flip these around into estimates for the class from $p(Y = c|\boldsymbol{X})$.

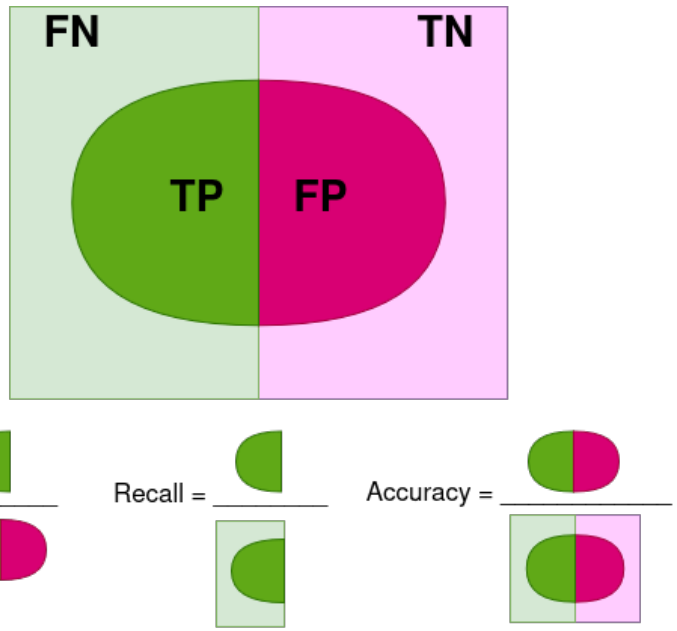
$$c^* = \underset{c}{\textbf{argmax}}\ \log p(Y = c|\ \boldsymbol{X};\ \boldsymbol{\beta}^*) \propto \log p(Y = c, \boldsymbol{X};\ \boldsymbol{\beta}^*)$$

- Eg. LDA, QDA, Naive Bayes.

Binary classification performance

Suppose a response variable (y) has two outcomes: positive or negative.

		Predicted samples	
		Predicted Positive	Predicted Negative
Actual samples / Ground truth	Actual Positive (P)	True Positive (TP)	False Negative (FN) $FN = P - TP$
	Actual Negative (N)	False Positive (FP)	True Negative (TN) $TN = N - FP$



- True positive (TP): correctly identified positive
- True negative (TN): correctly identified negative
- False positive (FP): incorrectly identified positive
- False negative (FN): incorrectly identified negative

Standard classification indices

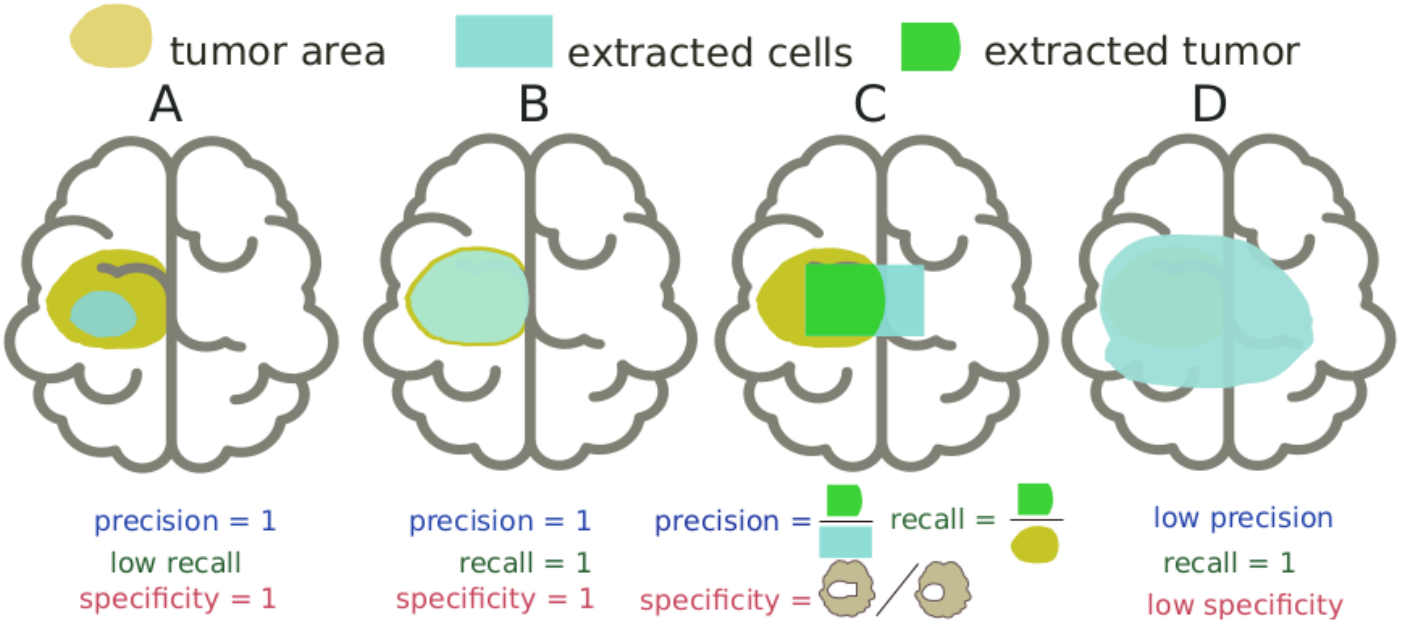
		Predicted samples			
		Predicted Positive	Predicted Negative		
Actual samples / Ground truth	Actual Positive (P)	**True Positive (TP)**	**False Negative (FN)** $FN = P - TP$	**Recall**, Sensitivity. True Positive Rate (TPR) $\frac{TP}{P}$	**Type II error**, False Negative Rate (FNR) $\frac{FN}{P} = 1 - TPR$
	Actual Negative (N)	**False Positive (FP)**	**True Negative (TN)** $TN = N - FP$	**Type I error**, False Positive Rate (FPR) $\frac{FP}{N}$	**Specificity**, Selectivity, True Negative Rate (TNR) $\frac{TN}{N} = 1 - FPR$
		Precision $\frac{TP}{TP + FP}$		**Accuracy** $\frac{TP + TN}{P + N}$	

F1

$$\frac{2Precision \cdot Recall}{Precision + Recall} = \frac{2TP}{2TP + FP + FN}$$

Trade-off in detecting brain tumor

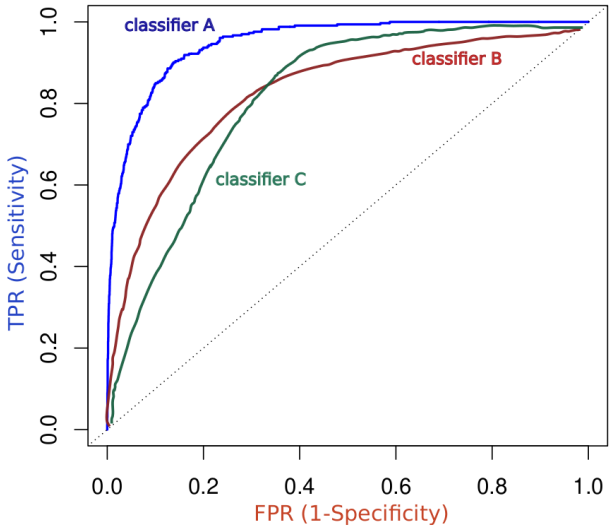
A neuro surgeon makes a decision to remove a brain tumor



- A conservative move (A): avoid to remove healthy cells by extracting only little.
- A bold move (D): all tumor must be gone, but unavoidably remove healthy cells.
- Always consider a combination of (sens,spec) or (precision,recall) – there is a price to pay; when one index increases, the other index would drop.

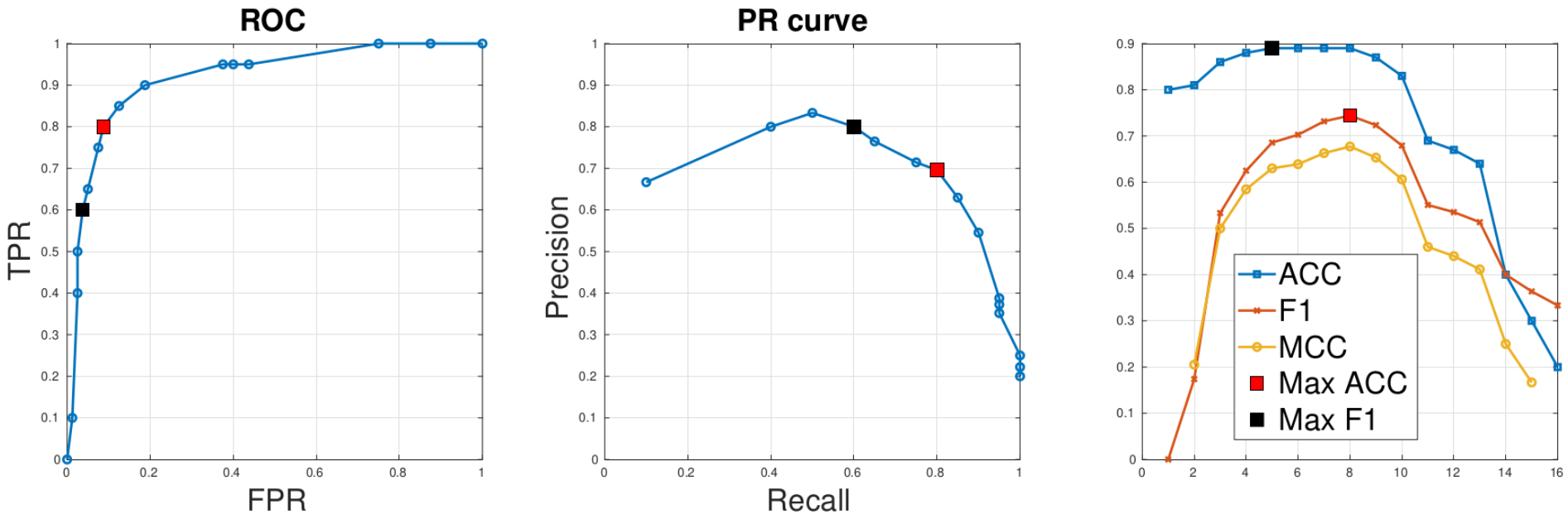
Receiver operating characteristic (ROC)

Receiver operating characteristic (ROC) is a plot between FPR (sensitivity/recall) and TPR (specificity).
Example: assign Y to the class '1' if $P(Y = 1|X = x) >$ threshold.



- Each point on ROC is associated with a **threshold/hyperparameter** of a classifier.
- A classifier performs better than a random guess, if ROC lies above the diagonal line.
- A better classifier has the ROC curve toward **the top left corner**.
- Overall performance is summarized over all possible thresholds is given by **area under curve (AUC)**.

Precision-recall curve



- **Black square** is the point for which ACC is maximum.
- **Red** is the point for which F1 is maximum.
- **Both black and red** points lie on the top-left corner of ROC, and on the top-right corner of PR (Precision-Recall) curve, indicating that these points are efficient.
- In practice, an operating point can be chosen by selecting the classifier's hyperparameter that maximizes some index (F1, MCC, ACC) on validated data.

Multi-class classification

Let's consider a C-class classification labeled as G1, G2, G3, G4, G5 ($C = 5$). The confusion matrix $\in \mathbb{R}^{C \times C}$ contains the number of samples in each predicted class is calculated according to the binary classification indices by picking a class of interest positive, and regard the remaining classes as negative.

Actual \ Predicted	G1	G2	G3	G4	G5	sum
G1	9	2	2	4	2	19
G2	1	10	2	2	5	20
G3	2	5	14	1	5	27
G4	1	5	2	8	3	19
G5	1	2	2	2	8	15
sum	14	24	22	17	23	100

Evaluation metrics to consider:

- Accuracy
- Precision, recall, F1 in *per-class performance*
- Precision, recall, F1 using *macro-averaging*
- ... ... *micro-averaging*
- ... ... *weighted-averaging*

Accuracy

A C-class classification labeled as G1, G2, G3, G4, G5 ($C = 5$).

$$\text{Accuracy} = \frac{\sum_c TP_c}{\text{all predictions}}$$

For the confusion matrix below, the accuracy is $\frac{9+10+14+8+8}{100} = 0.49$

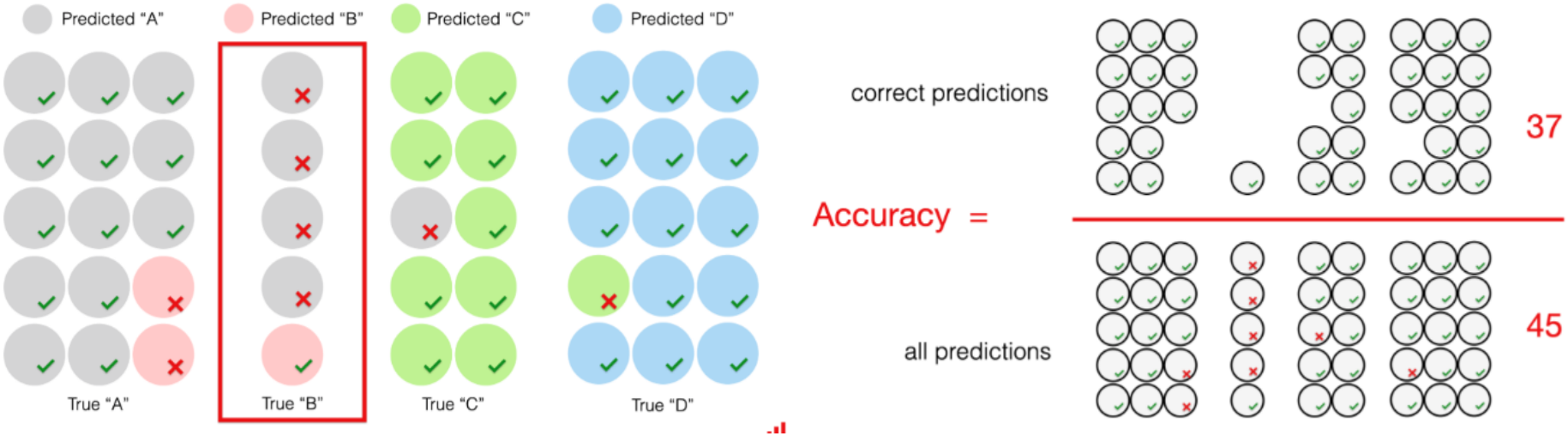
Actual \ Predicted	G1	G2	G3	G4	G5	sum
G1	9	2	2	4	2	19
G2	1	10	2	2	5	20
G3	2	5	14	1	5	27
G4	1	5	2	8	3	19
G5	1	2	2	2	8	15
sum	14	24	22	17	23	100

The good, the bad of accuracy for multi-classes.

- The good. Straightforward!
- The bad. Disregard class balance and the cost of different errors.

What is class balance and the cost of different errors?

Problem with accuracy



Let's look at an example from [1], the model did not do a very good job of predicting Class "B." However, since there were only 5 instances of this class, it did not impact the the overall accuracy is $37/45 = 82\%$!

Performance per class: confusion matrix of C-class

Performance per class.

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c} \quad \text{and} \quad \text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}$$

Then,

$$\text{F1}_c = 2\frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

From the confusion matrix example, we can calculate precision, recall, and F1 of class G3 as follows:

$$\text{Precision}_{\text{G3}} = \frac{14}{22} = 63.6\% \quad \text{Recall}_{\text{G3}} = \frac{14}{27} = 51.9\% \qquad \text{F1}_{\text{G3}} = 2\frac{\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}} = 57.1\%$$

		Predicted				
		G1	G2	G3	G4	G5
Actual	G1	9	2	2	4	2
	G2	1	10	2	2	5
	G3	2	5	14	1	5
	G4	1	5	2	8	3
	G5	1	2	2	2	8

G3 = positive	FP	FN	TP	TN

The good/the bad with per-class performance.

- The good...
  - Specific to classes of interest.
  - Helpful when dealing with imbalanced classes. Clearly expose issues in each class.
- The bad...The more classes, the more metrics you get.

Micro, Macro, and weighted-averaging

Three common ways summarize Precision and Recall across different classes as one value:

1. **Macro-averaging**
2. **Micro-averaging**
3. **Weighted-averaging**

Micro vs. Macro-averaging

1 **Macro-averaging** computes each precision$_c$ and then average.

$$\text{Precision} = \frac{1}{C}\sum_{c=1}^{C}\frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad \text{Recall} = \frac{1}{C}\sum_{c=1}^{C}\frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \quad \text{and} \quad \text{F1} = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

In fact, the above formulation can be re-written as ...

$$\text{Precision} = \frac{1}{C}\sum_{c=1}^{C}\text{Precision}_c \quad \text{and} \quad \text{Recall} = \frac{1}{C}\sum_{c=1}^{C}\text{Recall}_c.$$

- *Good ...*

  Useful when all classes are equally important, and you want to know how well the classifier performs on average across them.

  Useful when you have an imbalanced dataset and want to **ensure each class equally contributes** to the final evaluation.

- *Bad...* It can still distort the perception of performance; often this depends on the number of classes (C).

  Small C. It can make the classifier look "worse" due to low performance in an unimportant class.

  Large C. It can disguise poor performance in the critical minority class when the overall number of classes is large. In this case, **the "contribution" of each class is diluted**.

Micro vs. Macro-averaging

2 **Micro-averaging** takes a sum of all TP_c's and then compute precision/recall.

$$\text{Precision} = \frac{\sum_c \text{TP}_c}{\sum_c (\text{TP}_c + \text{FP}_c)}, \quad \text{Recall} = \frac{\sum_c \text{TP}_c}{\sum_c (\text{TP}_c + \text{FN}_c)}, \quad \text{and} \quad \text{F1} = \text{Precision} = \text{Recall}$$

Likewise, the above formulation can be re-written as ...

$$\text{Precision} = \frac{\text{TP}_1 + \text{TP}_2 + ... + \text{TP}_C}{\text{TP}_1 + \text{FP}_1 + \text{TP}_2 + \text{FP}_2 + ... + \text{TP}_C + \text{FP}_C}$$

$$\text{Recall} = \frac{\text{TP}_1 + \text{TP}_2 + ... + \text{TP}_C}{\text{TP}_1 + \text{FN}_1 + \text{TP}_2 + \text{FN}_2 + ... + \text{TP}_C + \text{FN}_C}$$

Note. Every FP for one class is an FN for another class. Thus, micro-average precision and recall results in the same number.

- *Good ...*

  Actually, Precision = Recall = Accuracy. Thus, it's advantages are similar to the accuracy's. Good when you want to account for the total number of misclassifications in the dataset. It gives equal weight to each instance and will have a higher score when the overall number of errors is low.

- *Bad...*

  Micro-averaging can overemphasize the performance of the majority class, especially when it dominates the dataset.

Weighted-averaging

3 **Weighted-averaging** takes the precision and recall by class and weights each of them with proportion of samples in each class.

$$\text{Precision} = \sum_{i=1}^{C} w_c \text{Precision}_c, \quad \text{Recall} = \sum_{i=1}^{C} w_c \text{Recall}_c, \quad \text{and} \quad \text{F1} = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- *Good ...*

  Actually, it is similar to macro-averaging if $w_c = \frac{1}{N}$ for all $c \in \{1, 2, ..., C\}$. You can actually set the weight representation w_c for each class to solve the problem of macro-averaging.

- *Bad...*

  There is no rule as to how each w_c should be set. If you want to emphasize on the class with small samples, you can construct w_c to be inversely proportional to the number of samples in each class.

Summary: micro, macro, and weighted-averaging

Three common ways summarize Precision and Recall across different classes as one value:

1 **Macro-averaging** computes each precision$_c$ and then average.

$$\text{Precision} = \frac{1}{C}\sum_{c=1}^{C}\frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad \text{Recall} = \frac{1}{C}\sum_{c=1}^{C}\frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \quad \text{and} \quad \text{F1} = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

2 **Micro-averaging** takes a sum of all TP$_c$'s and then compute precision/recall.

$$\text{Precision} = \frac{\sum_c \text{TP}_c}{\sum_c(\text{TP}_c + \text{FP}_c)}, \quad \text{Recall} = \frac{\sum_c \text{TP}_c}{\sum_c(\text{TP}_c + \text{FN}_c)}, \quad \text{and} \quad \text{F1} = \text{Precision} = \text{Recall}$$

Note. Every FP for one class is an FN for another class. Thus, micro-average precision and micro-average will result in the same number.

3 **Weighted-averaging** takes the precision and recall by class and weights each of them with proportion of samples in each class.

$$\text{Precision} = \sum_{i=1}^{C} w_c \text{Precision}_c, \quad \text{Recall} = \sum_{i=1}^{C} w_c \text{Recall}_c, \quad \text{and} \quad \text{F1} = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

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